WORKED SOLUTION

STEP III 2015, Question 10

Mechanics • Kinematics

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Read the question before you go on. It is in the STEP Support Programme's question database:

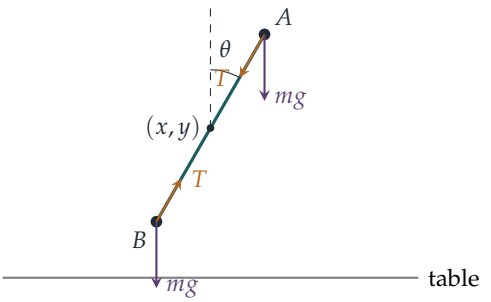
step.maths.org/questions/15-s3-q10

The key idea

Describe the system by the motion of the midpoint and the rotation of the rod. Write each particle's position in terms of x, y and θ, differentiate twice, and apply Newton's second law to each particle. Adding the two equations removes the tension and gives the motion of the centre of mass. Subtracting them and resolving perpendicular to the rod removes it again and leaves the rotation. Once all three accelerations are known, the final part is projectile motion plus steady rotation.

Solution

We model the two masses as particles joined by a light rigid rod. The only forces on each particle are its weight and the force from the rod, which acts along the rod. Let T be the tension, so $T > 0$ means the rod pulls each particle towards the midpoint. (A rod can also push, and then $T < 0$; nothing below depends on the sign.)



Equations of motion for particle A

Let A be the particle on the end of the rod that starts above the midpoint, so the vector from the midpoint to A is $a(\sin\theta, \cos\theta)$. Then A is at

$$\mathbf{r}_A = \begin{pmatrix} x + a\sin\theta \\ y + a\cos\theta \end{pmatrix}, \qquad \text{so} \qquad \dot{\mathbf{r}}_A = \begin{pmatrix} \dot{x} + a\dot{\theta}\cos\theta \\ \dot{y} - a\dot{\theta}\sin\theta \end{pmatrix}$$

by the chain rule. This is the required velocity. Differentiating again, using the product rule on $\dot{\theta}\cos\theta$ and $\dot{\theta}\sin\theta$,

$$\ddot{\mathbf{r}}_A = \begin{pmatrix} \ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta \\ \ddot{y} - a\ddot{\theta}\sin\theta - a\dot{\theta}^2\cos\theta \end{pmatrix}.$$

The rod pulls A towards the midpoint, in the direction $-(\sin\theta, \cos\theta)$, with force of size T. Gravity contributes $-mg(0,1)$. So Newton's second law for A gives exactly the stated equation:

$$m\begin{pmatrix} \ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta \\ \ddot{y} - a\ddot{\theta}\sin\theta - a\dot{\theta}^2\cos\theta \end{pmatrix} = -T\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} - mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{A}$$

The other particle

For B, at the other end, replace a by $-a$ throughout. The rod now pulls B in the direction $+(\sin\theta, \cos\theta)$:

$$m\begin{pmatrix} \ddot{x} - a\ddot{\theta}\cos\theta + a\dot{\theta}^2\sin\theta \\ \ddot{y} + a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta \end{pmatrix} = T\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} - mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{B}$$

Here B has velocity $(\dot{x} - a\dot{\theta}\cos\theta,\ \dot{y} + a\dot{\theta}\sin\theta)$.

Deducing the accelerations

Adding (A) and (B) cancels every term containing a or T:

$$2m\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = -2mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \text{so} \qquad \ddot{x} = 0, \quad \ddot{y} = -g.$$

Subtracting (B) from (A) and dividing by 2:

$$\begin{aligned} ma\ddot{\theta}\cos\theta - ma\dot{\theta}^2\sin\theta &= -T\sin\theta, \\ -ma\ddot{\theta}\sin\theta - ma\dot{\theta}^2\cos\theta &= -T\cos\theta. \end{aligned}$$

Ask yourself How do we get rid of T, which we don't know?

The tension acts along the rod, so we resolve perpendicular to it, in the direction $(\cos\theta, -\sin\theta)$. That means taking $\cos\theta$ times the first equation minus $\sin\theta$ times the

second. The right-hand sides give $-T\sin\theta\cos\theta + T\cos\theta\sin\theta = 0$, and the $\dot\theta^2$ terms cancel in the same way. What is left is

$$ma\ddot\theta(\cos^2\theta + \sin^2\theta) = 0,$$

and since $m, a \neq 0$ this gives $\ddot\theta = 0$. (Resolving along the rod instead gives $T = ma\dot\theta^2$, the tension needed for circular motion about the midpoint.)

The final part

At $t = 0$ the rod is vertical, so $\theta = 0$. Particle A is then the higher one, with velocity $(\dot x + a\dot\theta,\ \dot y)$, and B is the lower one, with velocity $(\dot x - a\dot\theta,\ \dot y)$. Matching these with the given initial velocities:

$$\dot x + a\dot\theta = u, \qquad \dot x - a\dot\theta = 0, \qquad \dot y = v,$$

so initially $\dot x = \frac{1}{2}u$, $\dot\theta = \dfrac{u}{2a}$ and $\dot y = v$.

By the previous part, $\dot\theta$ stays constant, so $\theta = \dfrac{ut}{2a}$. We take $u > 0$, so that θ increases. Also $\ddot y = -g$ with $y(0) = h$, so

$$y = h + vt - \tfrac{1}{2}gt^2.$$

The rod has turned through $\frac{1}{2}\pi$ when $\dfrac{ut}{2a} = \dfrac{\pi}{2}$, that is, at $t = \dfrac{\pi a}{u}$. The rod is then horizontal, so both particles are at the height y of the midpoint. They both hit the table at that moment exactly when $y = 0$ there:

$$h + v\cdot\frac{\pi a}{u} - \frac{1}{2}g\cdot\frac{\pi^2a^2}{u^2} = 0.$$

Multiplying by $2u^2$ and rearranging gives

$$2hu^2 = \pi^2a^2g - 2\pi uva,$$

as required.

Checks

- With $v = 0$, the time to fall is $\sqrt{2h/g}$, and this must equal $\pi a/u$. Squaring gives $2hu^2 = \pi^2a^2g$, which matches.
- Units: every term in the result has dimensions $\mathrm{L}^3\mathrm{T}^{-2}$.
- We simulated two particles joined by a rigid rod on a computer, for several random choices of a, u, v with $g = 9.8$ and h given by the formula. For example, $a \approx 1.754$ gives $h \approx 17.93$. In every case both particles reached the table together at $t = \pi a/u$, to within 10^{-10}, with the rod horizontal.

Common slips

- Getting the direction of the tension wrong for one of the particles. The rod pulls each particle towards the midpoint, so the two forces are equal and opposite.
- Using a different value of T in each equation. The rod is light, so the net force on it must be zero, and the tension is the same at both ends.
- Forgetting the $-a\dot{\theta}^2$ terms when differentiating $\dot{\theta}\cos\theta$. Without them the subtraction step does not come out cleanly.

Beyond STEP

This question is a small instance of a general theorem about systems of particles. The centre of mass moves as if all the mass and all the external force were concentrated there. Internal forces cancel in pairs, so with no external torque about the centre of mass the angular momentum about it is conserved. In first-year mechanics you would also meet the Lagrangian approach, which takes x, y, θ as generalised coordinates and never needs T at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers the centre of mass and rigid-body rotation thoroughly.

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